

Fernando Fernández Andueza

Gameplay & AI Programmer

anduezadev@gmail.com · [linkedin.com/in/fernando-fernandez-andueza](https://www.linkedin.com/in/fernando-fernandez-andueza) · anduezadev.itch.io

Valencia, Spain · EU citizen · open to relocation within Spain and to remote work · Spanish (native)

PROFILE

Gameplay and AI programmer. I built the enemy AI for Beerserker: The Stolen Brew (Steam, Early Access) with StateTree in Unreal Engine 5. I built XEMA, a custom C++ engine, from scratch, with OpenGL 4.5, deferred PBR rendering and an ECS core. I ported it to Nintendo Switch, where I have only tested it on an emulator, and a PlayStation 4 port is under way: the ECS and simple-geometry rendering work, on a devkit. Before that I spent two and a half years as a fullstack developer on ERP software.

EXPERIENCE

Gameplay & AI Programmer · Beerserker: The Stolen Brew (store.steampowered.com/app/4522440) · Final-year project at ESAT. Developed by Grumpy Games and ESAT, published on Steam by ESAT (Early Access, free)

Sept 2025 – Jul 2026

About 20 people across programming, art, design, music and production. I implemented the four enemy archetypes in StateTree, in C++ and Blueprints, with custom tasks and conditions, and used EQS to query the environment. I also built a templated object pool as a plugin (enemies, projectiles, decals and particles), so it is modular and can be taken to other projects, and the enemy spawners, and took part in designing the architecture of a custom GAS-based action system. I was responsible for configuring and uploading the Steam builds.

Gameplay Programmer · Roadtrip: The Engine of Madness (store.steampowered.com/app/3776540) · Oniric Tales

Dec 2024 – Jun 2025

Design, architecture and development of core gameplay systems. Built a scalable component-based dialogue system, an AI perception system integrated into behaviour trees, and a radio communication system; contributed to the Behavior Graph, general gameplay mechanics, profiling and bug fixing.

Fullstack Developer · AHORA

Sept 2020 – Mar 2023

Frontend of an enterprise management system with Vue.js, TypeScript, SCSS, HTML, Bootstrap, Flexbox and DevExtreme. Developed products and management systems with FlexyGo, an in-house low-code framework, and built components and extensions for it. Backend in VB and C# on MSSQL; frontend with TypeScript, HTML, CSS and WebComponents. Handled and resolved client incidents, with frequent travel to the client's site. Worked with Agile methodologies (Scrum).

SELECTED PROJECTS

XEMA — custom 3D engine (andueza013.github.io/PortfolioGameDev/graphic-engine)

C++ · OpenGL 4.5 · Nintendo Switch (emulator)

A graphics engine I built entirely myself in C++ and OpenGL. It has an ECS architecture, deferred PBR rendering with shadow mapping and instanced rendering, a JobSystem for tasks such as asynchronous mesh loading, Lua scripting and PhysX for physics. I use GLFW for the window and input, and ImGui for visual markers of entities, components and performance. It runs on PC and I have ported it to Nintendo Switch, where I have only tested it on an emulator. The PS4 port is in progress: so far the ECS and rendering of simple geometry work, on a devkit. Mesh loading, textures, instancing, lights, shadows and PBR are still missing.

Abandoned Hospital

Unreal Engine 5 · C++ · AI Perception, EQS

Class exercise to work with AI Perception and EQS in Unreal Engine, built with a behaviour tree, for enemy and NPC behaviour. Once I had handed it in, I rebuilt it on my own with free assets and rewrote the AI from scratch with StateTree.

EDUCATION & CERTIFICATIONS

HND in Video Game Programming · ESAT — Escuela Superior de Arte y Tecnología · Valencia 2023 – 2026

Specialisation in video game programming, especially in C++. Over three years I have built a graphics engine from scratch, which taught me how graphics engines work under the hood, both in software architecture and in the graphics pipeline, and published a game on Steam with Unreal Engine 5. I have also worked on procedural generation, implementing WFC (in my own engine); a templated FSM from scratch; Markov chains (to choose which attack the Beerserker tank uses, with weights that adjust dynamically); A* pathfinding; cellular automata and L-systems. During the three years we also learned to program in assembly and to read the assembly our C++ compiles to, in order to identify bottlenecks and performance problems, especially in inner loops, and fix them with optimisation techniques. This last year I have specialised as an AI programmer, building the four enemy archetypes with StateTree in UE 5.6.

DAM — Multiplatform Application Development · IES El Grao · Valencia 2018 – 2020

Java programming with Eclipse and NetBeans for desktop, and Android Studio for mobile. Relational databases, HTML and CSS. Fundamentals of OOP and version control.

Scrum Master & Product Owner · Scrum.org / International Agile Institutes Professional credentials

Certified in both roles: bridging design requirements and system architecture, optimising production pipelines and managing technical debt across team iterations.

TECHNICAL SKILLS

Game engines	Unreal Engine 5 (C++ & Blueprints), Unity (C#), XEMA — custom C++ engine
AI & gameplay	StateTree, Behaviour Trees, EQS, AI Perception, Animation systems, Object pooling
Languages	C++, C#, C, Lua
Graphics & systems	OpenGL 4.5 DSA, Deferred shading / PBR, Vertex & fragment shaders, ECS architecture, ImGui, GLFW, Multithreading & job systems, Memory management & pointers, PhysX 5
Debugging & profiling	Unreal Insights, Visual Studio debugger, RenderDoc
Platforms	PC (Windows), Nintendo Switch (emulator), PlayStation 4 (partial, devkit)
Tooling	Git, Perforce, Plastic SCM, Premake5
Outside games	TypeScript / JavaScript, HTML / CSS, Vue.js, SQL Server, Java, VB .NET

Last updated: September 2026